REP-050 · PLAN-020 built: rules version 5 (lorry every day, no second-half walk-in)

Game	Fruit Market (rules version 5 on local branch `workflow/plan-020`; preview and live unchanged)
Report date	2026-10-27 23:35 (Asia/Kuala_Lumpur)
Type	Plan implementation. **Game change, local only, not pushed, not on the preview**
Request	"why is the number in decimal? remember that rules should be able to be implemented on a tabletop elsewhere, else do plan a + c, target is so that the board would not be empty / ppl cant buy anything"
Plan	PLAN-020 (approved; implemented; one check NOT RUN)

STATUS

Rules version 5 is built on local branch `workflow/plan-020`, two commits on top of the records branch. Not pushed, so CI has not run and the preview is unchanged. Lint, types, 92 unit tests and the build pass. The browser tests (`verify:full`) did not run.

SUMMARY

Option A + C is in: the lorry comes every day, and walk-ins are players minus 2 every day (plus 1 on a Market Day). In simulation the market is no longer empty (0.0% to 1.0% of shop turns, down from up to 22%), and late-game demand is level with early-game. Both rules are printed numbers, playable by hand, so the tabletop master (`RULES.md`) is updated to match.

PLAN OR WORK COMPLETED

How I read the message. "else do plan a + c" = go with A + C. "target is so that the board would not be empty" = the target is that shops can always buy something, measured as the share of shop turns that find an empty market. "Why the number in decimal" is answered below.

Why the numbers had decimals. They are averages over hundreds of simulated games. A single game always has whole numbers (for example 5 customers on a day); the average of 300 games is 5.3. The percentages are shares of shop turns.

Changes

- `src/engine/game.ts`: the lorry card is drawn every day; walk-ins are players minus 2, plus 1 on a Market Day, plus any event.
- `src/engine/ai.ts`: computer shops no longer treat a Market Day as a no-lorry shortage (one fixed weight for scarce fruit).
- `src/net/protocol.ts`: rules version 5 (old saves will not resume; daily boards are kept per version; online players need the same version).
- Screens: the lorry shows on Market Days, the strip says "Market Day: one more walk-in", and How to play, set-up and hint text are updated.

- `RULES.md` (tabletop master) updated: lorry every day, walk-ins every day, version 5, balance note.
- Tests updated: walk-ins, Market Day delivery, demand staying level, version 5.

Tabletop check (DEC-027). The farmer card is flipped every day, as it already was on other days. Walk-ins are a printed number: players minus 2. Nothing new to move or count.

Results after the change (computer players, all Normal, same seeds 1 to 300)

Shops	Customers/day, first 3 / last 3 days	Fruit sold vs wanted, first / last	Shop turns short of their buy limit, first / last	Shop turns with an empty market, first / last
2	2.1 / 2.2	63% / 86%	0.5% / 0.1%	0.1% / 0.0%
3	3.5 / 3.7	60% / 79%	1.4% / 0.6%	0.3% / 0.0%
4	4.9 / 5.3	62% / 75%	1.5% / 2.4%	0.6% / 0.2%
4 (Quick, 8 days)	4.9 / 5.5	62% / 73%	1.5% / 4.2%	0.6% / 1.0%

Before the change (4 shops, 10 days): last 3 days had customers 8.9, fill 30%, turns short 35%, empty-market turns 22%. The highest empty-market rate on any single day after the change is 1.6% (4 shops, day 7).

VALIDATION

Check	Result	Evidence
Owner target: market not empty	PASS (simulation)	0.0% to 1.0% of shop turns find an empty market; computer players only
Lint	PASS	`npx eslint .` no output
Type checks (app and server)	PASS	`tsc --noEmit` and `tsc -p server --noEmit`
Unit tests	PASS	92 of 92 (`npx vitest run`)
Build	PASS	`npm run build`
Records check	PASS	Run with a patched copy; the repo script fails on Windows paths
Browser tests (`verify:full`)	NOT RUN	Needs Playwright browsers; about 15 minutes
Human playtest of rules 5	NOT RUN	None; computer players only
Tabletop paper test	NOT RUN	Not done; hand-played ageing is still unproven

CHANGELOG

Added: none. Changed: lorry every day; no extra walk-in in the second half; computer shops stop treating Market Days as a shortage; text updated; rules version 5. Removed: the no-lorry Market

Day. Player impact: the market no longer runs empty mid-game and the last days are no longer flooded. State: implemented, unreleased, local only.

RISKS AND COSTS

- **2-shop games are now quiet:** about 2 customers a day all game (they were 4.9 in the last days before). The market is well supplied (fill 86%), so shops sell most of what they put out but earn less per day. Worth a look in a human game; a 2-shop tweak could come later.
- **Market Days lost their tension.** They are now only "one more walk-in" plus the speciality draft afterwards.
- **Computer players only.** Humans may buy and price differently.
- Not pushed, so CI has not checked it and the preview does not show it yet.
- Old saves cannot be resumed and the daily leaderboard restarts for version 5; online rooms need matching versions.
- Cost: RM0.

NEXT ACTION OR APPROVAL

The owner decides whether to push `workflow/plan-020` to GitHub (CI runs; main stays untouched), and later whether to merge it to `main` (the preview updates after CI passes). Also open: PLAN-021 time target, PLAN-023 design choice, and whether to approve PLAN-022.

DOCUMENT RECORD

Report REP-050. DEC-029 approved (owner words recorded); PLAN-020 implemented, T1 and T2 PASS, T3 NOT RUN (browser tests), T4 NOT RUN. Register: `docs/workflow/DECISIONS.md`. **PDF EXPORT BLOCKED:** pandoc is not installed on this machine (the repo's dependencies now are); recovery: install pandoc, then run `npm run records` (needs owner approval).